

CS3300 - Compiler Design

Parsing

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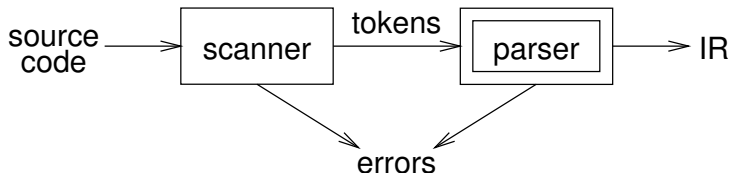
IIT Madras

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Where we are

- 1 Context-free grammars
 - Grammars and notation
 - Derivations and parse trees
 - Ambiguity
- 2 Top-down parsing

The role of the parser



A parser

- performs context-free syntax analysis
- guides context-sensitive analysis
- constructs an intermediate representation
- produces meaningful error messages
- attempts error correction

For the next several classes, we will look at parser construction

Syntax analysis by using a CFG

Context-free syntax is specified with a *context-free grammar*.

Formally, a CFG G is a 4-tuple (V_t, V_n, S, P) , where:

- V_t is the set of *terminal* symbols in the grammar.

- For our purposes, V_t is the set of tokens returned by the scanner.

- V_n , the *nonterminals*, is a set of syntactic variables that denote sets of (sub)strings occurring in the language.

- S is a distinguished nonterminal ($S \in V_n$) denoting the entire set of strings in $L(G)$.

- This is sometimes called a *goal symbol*.

- P is a finite set of *productions*

- Each production must have a single non-terminal on its left hand side.

The set $V = V_t \cup V_n$ is called the *vocabulary* of G .

Notation and terminology

- $a, b, c, \dots \in V_t$
- $A, B, C, \dots \in V_n$
- $U, V, W, \dots \in V$
- $\alpha, \beta, \gamma, \dots \in V^*$
- $u, v, w, \dots \in V_t^*$

If $A \rightarrow \gamma$ then $\alpha A \beta \Rightarrow \alpha \gamma \beta$ is a *single-step derivation* using $A \rightarrow \gamma$

Similarly, $\Rightarrow^*$ and $\Rightarrow^+$ denote derivations of ≥ 0 and ≥ 1 steps

If $S \Rightarrow^* \beta$ then β is said to be a *sentential form* of G

$L(G) = \{w \in V_t^* \mid S \Rightarrow^+ w\}$, $w \in L(G)$ is called a *sentence* of G

Note, $L(G) = \{\beta \in V^* \mid S \Rightarrow^* \beta\} \cap V_t^*$

Syntax analysis

Grammars are often written in Backus-Naur form (BNF).

Example:

1		$\langle \text{goal} \rangle ::= \langle \text{expr} \rangle$
2		$\langle \text{expr} \rangle ::= \langle \text{expr} \rangle \langle \text{op} \rangle \langle \text{expr} \rangle$
3		num
4		id
5		$\langle \text{op} \rangle ::= +$
6		-
7		*
8		/

This describes simple expressions over numbers and identifiers.

In a BNF for a grammar, we represent

- ① non-terminals with angle brackets or capital letters
- ② terminals with `typewriter` font or underline
- ③ productions as in the example

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Derivations

We can view the productions of a CFG as rewriting rules.
Using our example CFG (for $x + 2 * y$):

$$\begin{aligned}\langle \text{goal} \rangle &\Rightarrow \langle \text{expr} \rangle \\ &\Rightarrow \langle \text{expr} \rangle \langle \text{op} \rangle \langle \text{expr} \rangle \\ &\Rightarrow \langle \text{id}, x \rangle \langle \text{op} \rangle \langle \text{expr} \rangle \\ &\Rightarrow \langle \text{id}, x \rangle + \langle \text{expr} \rangle \\ &\Rightarrow \langle \text{id}, x \rangle + \langle \text{expr} \rangle \langle \text{op} \rangle \langle \text{expr} \rangle \\ &\Rightarrow \langle \text{id}, x \rangle + \langle \text{num}, 2 \rangle \langle \text{op} \rangle \langle \text{expr} \rangle \\ &\Rightarrow \langle \text{id}, x \rangle + \langle \text{num}, 2 \rangle * \langle \text{expr} \rangle \\ &\Rightarrow \langle \text{id}, x \rangle + \langle \text{num}, 2 \rangle * \langle \text{id}, y \rangle\end{aligned}$$

We have derived the sentence $x + 2 * y$.

We denote this $\langle \text{goal} \rangle \Rightarrow^* \text{id} + \text{num} * \text{id}$.

Such a sequence of rewrites is a *derivation* or a *parse*.

The process of discovering a derivation is called *parsing*.

Derivations

At each step, we chose a non-terminal to replace.

This choice can lead to different derivations.

Two are of particular interest:

leftmost derivation

the leftmost non-terminal is replaced at each step

rightmost derivation

the rightmost non-terminal is replaced at each step

The previous example was a leftmost derivation.

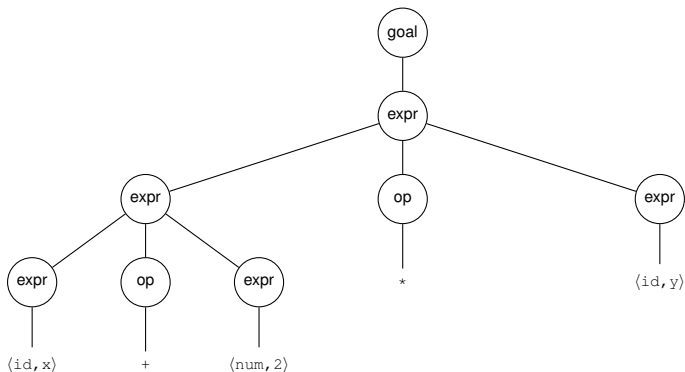
Rightmost derivation

For the string $x + 2 * y$:

$$\begin{aligned}\langle \text{goal} \rangle &\Rightarrow \langle \text{expr} \rangle \\ &\Rightarrow \langle \text{expr} \rangle \langle \text{op} \rangle \langle \text{expr} \rangle \\ &\Rightarrow \langle \text{expr} \rangle \langle \text{op} \rangle \langle \text{id}, y \rangle \\ &\Rightarrow \langle \text{expr} \rangle * \langle \text{id}, y \rangle \\ &\Rightarrow \langle \text{expr} \rangle \langle \text{op} \rangle \langle \text{expr} \rangle * \langle \text{id}, y \rangle \\ &\Rightarrow \langle \text{expr} \rangle \langle \text{op} \rangle \langle \text{num}, 2 \rangle * \langle \text{id}, y \rangle \\ &\Rightarrow \langle \text{expr} \rangle + \langle \text{num}, 2 \rangle * \langle \text{id}, y \rangle \\ &\Rightarrow \langle \text{id}, x \rangle + \langle \text{num}, 2 \rangle * \langle \text{id}, y \rangle\end{aligned}$$

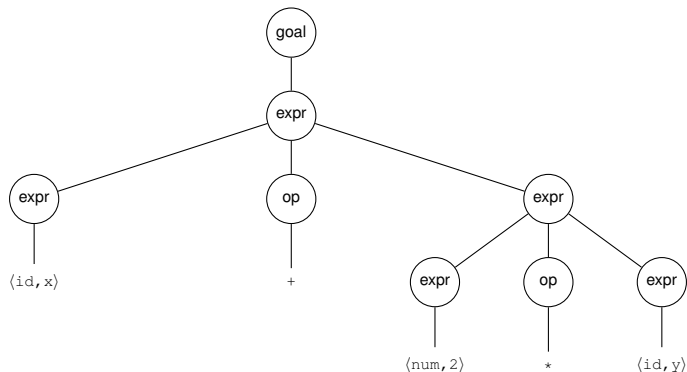
Again, $\langle \text{goal} \rangle \Rightarrow^* \text{id} + \text{num} * \text{id}$.

Precedence: the tree from the *rightmost* derivation



*Treewalk evaluation computes $(x + 2) * y$*
— the “wrong” answer!
Should be $x + (2 * y)$

Precedence: the tree from the *leftmost* derivation



Treewalk evaluation computes $x + (2 * y)$

— the answer we wanted.

Same grammar, same string, two different trees.

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Precedence

These two derivations point out a problem with the grammar.

It has no notion of precedence, or implied order of evaluation.

The grammar is ambiguous, as a string in the language can have multiple parse trees.

Is precedence the only source of ambiguity? Other examples of strings with multiple parse trees?

Ambiguity - Associativity

The expression $a-b-c$ may be parsed as:

- $(a-b) - c$ or
- $a - (b-c)$

In C, assignment $=$ is right-associative. $a=b=c$ may be parsed as:

- $a = (b=c)$ or
- $(a=b) = c$

Removing Ambiguity

To remove ambiguity, the grammar needs to be modified:

1		$\langle \text{goal} \rangle$	$::=$	$\langle \text{expr} \rangle$
2		$\langle \text{expr} \rangle$	$::=$	$\langle \text{expr} \rangle + \langle \text{term} \rangle$
3				$\langle \text{expr} \rangle - \langle \text{term} \rangle$
4				$\langle \text{term} \rangle$
5		$\langle \text{term} \rangle$	$::=$	$\langle \text{term} \rangle * \langle \text{factor} \rangle$
6				$\langle \text{term} \rangle / \langle \text{factor} \rangle$
7				$\langle \text{factor} \rangle$
8		$\langle \text{factor} \rangle$	$::=$	num
9				id

This grammar enforces a *precedence* and *associativity* on the derivation:

- terms *must* be derived from expressions
- forces the “correct” tree

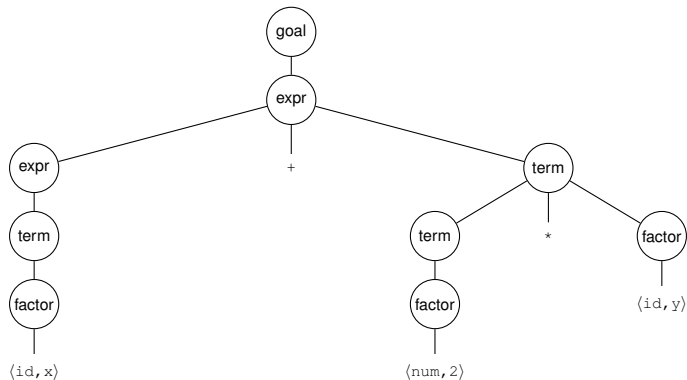
Precedence

Now, for the string $x + 2 * y$:

$$\begin{aligned}\langle \text{goal} \rangle &\Rightarrow \langle \text{expr} \rangle \\ &\Rightarrow \langle \text{expr} \rangle + \langle \text{term} \rangle \\ &\Rightarrow \langle \text{expr} \rangle + \langle \text{term} \rangle * \langle \text{factor} \rangle \\ &\Rightarrow \langle \text{expr} \rangle + \langle \text{term} \rangle * \langle \text{id}, y \rangle \\ &\Rightarrow \langle \text{expr} \rangle + \langle \text{factor} \rangle * \langle \text{id}, y \rangle \\ &\Rightarrow \langle \text{expr} \rangle + \langle \text{num}, 2 \rangle * \langle \text{id}, y \rangle \\ &\Rightarrow \langle \text{term} \rangle + \langle \text{num}, 2 \rangle * \langle \text{id}, y \rangle \\ &\Rightarrow \langle \text{factor} \rangle + \langle \text{num}, 2 \rangle * \langle \text{id}, y \rangle \\ &\Rightarrow \langle \text{id}, x \rangle + \langle \text{num}, 2 \rangle * \langle \text{id}, y \rangle\end{aligned}$$

Again, $\langle \text{goal} \rangle \Rightarrow^* \text{id} + \text{num} * \text{id}$, but this time, we build the desired tree.

Precedence



*Treewalk evaluation computes $x + (2 * y)$*

Role of CFGs in Compilers

CFGs offer significant advantages for language designers, compiler developers, and end-users of the compiler:

- A grammar gives a formal, precise, yet easy-to-understand syntactic specification of the programming languages. **Useful for end-users**
- For certain classes of grammars, there are procedures to automatically construct efficient parsers from the grammar description. **Useful for compiler developers**
- A grammar can reveal syntactic ambiguities and trouble spots. **Useful for language designers**
- A grammar imparts structure to a program, which is directly used for its translation into object code. **Useful for compiler developers**
- A grammar allows a language to be evolved iteratively by adding new constructs. **Useful for language designers and compiler developers**

Ambiguity

If a grammar has more than one **parse tree** for a single sentence, then it is *ambiguous*

(equivalently: more than one *leftmost* derivation)

Example:

```
<stmt> ::=  if <expr> then <stmt>
           |  if <expr> then <stmt> else <stmt>
           |  other
```

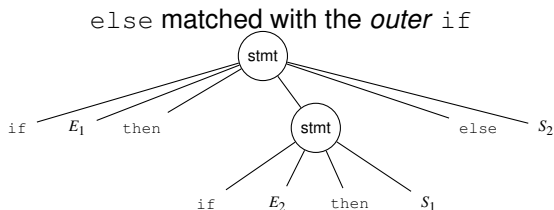
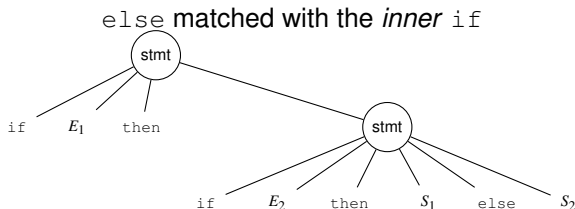
Consider deriving the sentential form:

if E_1 then if E_2 then S_1 else S_2

This is the classic **dangling else** problem.

The ambiguity is *purely grammatical*: it is visible in the productions alone, with no reference to types or declarations. A *context-free* ambiguity.

Ambiguity: two trees for the same string



Both yield if E₁ then if E₂ then S₁ else S₂. **Which if owns the else?**

Ambiguity

We would like to parse `if-then-else` statements using the following rule:

match each `else` with the closest unmatched `then`

Grammar which eliminates the ambiguity by following the above rule:

$\langle \text{stmt} \rangle$	$::=$	$\langle \text{matched} \rangle$
	$ $	$\langle \text{unmatched} \rangle$
$\langle \text{matched} \rangle$	$::=$	<code>if</code> $\langle \text{expr} \rangle$ <code>then</code> $\langle \text{matched} \rangle$ <code>else</code> $\langle \text{matched} \rangle$
	$ $	<code>other</code>
$\langle \text{unmatched} \rangle$	$::=$	<code>if</code> $\langle \text{expr} \rangle$ <code>then</code> $\langle \text{stmt} \rangle$
	$ $	<code>if</code> $\langle \text{expr} \rangle$ <code>then</code> $\langle \text{matched} \rangle$ <code>else</code> $\langle \text{unmatched} \rangle$

Did the rewrite lose anything?

Not a feeling. The precise question: do the two grammars generate the **same language**?

Here, yes. Both grammars are small, and it can be proved. A proof checked by Lean 4 sits next to this deck in the course repository:

https://github.com/fplaunchpad/cs3300_m26/tree/main/lectures/03_parsing

In general, whether two CFGs generate the same language is **undecidable**.

That means no algorithm answers it for *every* input. It does not mean any particular case is hard, and it does not mean there is no proof.

Ambiguity

Ambiguity is often due to confusion in the context-free specification.

Context-sensitive confusions can arise from *overloading*.

Example:

$$a = b + c$$

In many languages, $+$ can refer to both integer addition and floating point addition.

Disambiguating this statement requires context:

- need *values* of declarations
- not *context-free*
- really an issue of *type*

Rather than complicate parsing, we will handle this separately.

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- From scanning to parsing
- Left recursion
- Recursive descent and predictive parsing
- FIRST, FOLLOW and LL(1) grammars
- Building and using the LL(1) table
- When LL(1) fails

Scanning vs. parsing

Where do we draw the line?

$$\begin{aligned}\langle id \rangle &::= [a - zA - z]([a - zA - z] \mid [0 - 9])^* \\ \langle num \rangle &::= 0 \mid [1 - 9][0 - 9]^* \\ \langle op \rangle &::= + \mid - \mid * \mid / \\ \langle expr \rangle &::= \langle expr \rangle \langle op \rangle \langle expr \rangle \mid \langle id \rangle \mid \langle digit \rangle\end{aligned}$$

Regular expressions are used to classify:

- identifiers, numbers, keywords
- REs are more concise and simpler for tokens than a grammar
- more efficient scanners can be built from REs (DFAs) than grammars

Scanning vs. parsing

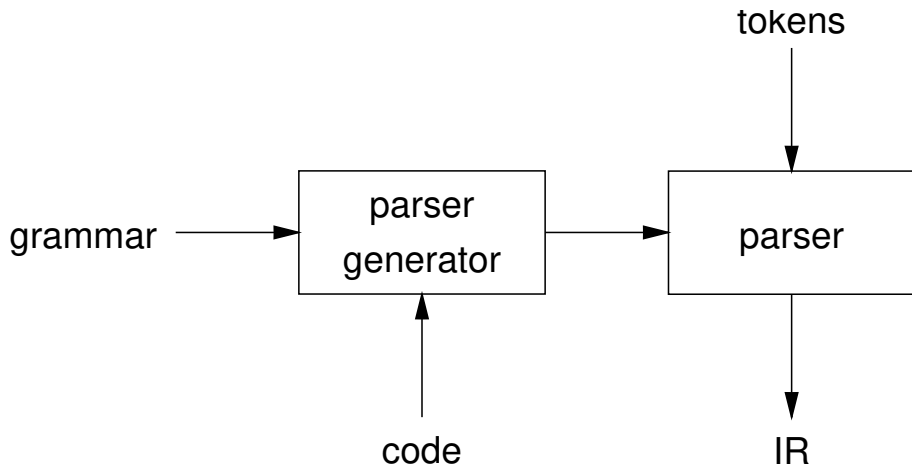
Context-free grammars are used to count:

- brackets: `()`, `begin...end`, `if...then...else`
- imparting structure
 - arithmetic expressions can be described by regular expressions
 - but, must deal with precedence and associativity separately ...

Syntactic analysis is complicated enough: grammar for C has around 200 productions.

Factoring out lexical analysis as a separate phase makes compiler more manageable.

Parsing: the big picture



Our goal is a flexible parser generator system

Different ways of parsing: Top-down Vs Bottom-up

Top-down parsers

- start at the root of derivation tree and fill in
- picks a production and tries to match the input
- may require backtracking
- some grammars are backtrack-free (*predictive*)

Bottom-up parsers

- start at the leaves and fill in
- start in a state valid for legal first tokens
- as input is consumed, change state to encode possibilities (*recognize valid prefixes*)
- use a stack to store both state and sentential forms

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Top-down parsing

A top-down parser starts with the root of the parse tree, labelled with the start or goal symbol of the grammar.

To build a parse, it repeats the following steps until the fringe of the parse tree matches the input string

- ① At a node labelled A , select a production $A \rightarrow \alpha$ and construct the appropriate child for each symbol of α
- ② When a terminal is added to the fringe that doesn't match the input string, backtrack
- ③ Find next node to be expanded (must have a label in V_n)

The key is selecting the right production in step 1.

Example

$$\langle term \rangle ::= id \mid num$$
$$\langle op \rangle ::= + \mid -$$
$$\langle expr \rangle ::= \langle expr \rangle \langle op \rangle \langle term \rangle \mid \langle term \rangle$$

Consider the string `x+5`.

Example: parsing $x+5$

$\langle term \rangle ::= id \mid num$

$\langle op \rangle ::= + \mid -$

$\langle expr \rangle ::= \langle expr \rangle \langle op \rangle \langle term \rangle$
 $\mid \langle term \rangle$

Input: $x + 5$

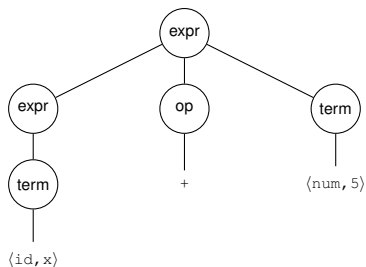
① $\langle expr \rangle ::= \langle expr \rangle \langle op \rangle \langle term \rangle$

② $\langle expr \rangle ::= \langle term \rangle$

③ $\langle term \rangle ::= id$, match x

④ $\langle op \rangle ::= +$, match $+$

⑤ $\langle term \rangle ::= num$, match 5



Example: the other choice at step 2

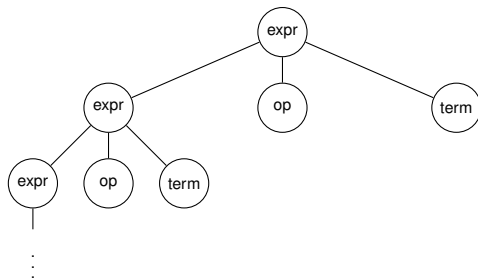
Step 2 could equally have taken

$\langle expr \rangle ::= \langle expr \rangle \langle op \rangle \langle term \rangle$ a second time.

Each application puts $\langle expr \rangle$ back at the left end of the fringe.

So no terminal ever reaches the front, no input is consumed, and the leftmost leaf in V_n is always $\langle expr \rangle$.

Nothing is matched, so nothing fails, so step 2 never backtracks.



Immediate Left-recursion

Top-down parsers cannot handle left-recursion in a grammar.

Formally, a grammar is *left-recursive* if

$\exists A \in V_n$ such that $A \Rightarrow^+ A\alpha$ for some string α

A grammar is said to be *immediate left-recursive* if

$\exists A \in V_n$ such that $A \rightarrow A\alpha$ for some string α

Our simple expression grammar is immediate left-recursive.

Eliminating immediate left-recursion

To remove immediate left-recursion, we can transform the grammar
Consider the grammar fragment:

$$\begin{array}{lcl} \langle \text{foo} \rangle & ::= & \langle \text{foo} \rangle \alpha \\ & | & \beta \end{array}$$

where α and β do not start with $\langle \text{foo} \rangle$

We can rewrite this as:

$$\begin{array}{lcl} \langle \text{foo} \rangle & ::= & \beta \langle \text{bar} \rangle \\ \langle \text{bar} \rangle & ::= & \alpha \langle \text{bar} \rangle \\ & | & \epsilon \end{array}$$

where $\langle \text{bar} \rangle$ is a new non-terminal

This fragment contains no immediate left-recursion

Eliminating immediate left-recursion

In general, if the grammar contains the following production rules:

$$\langle A \rangle ::= \langle A \rangle \alpha_1 \mid \langle A \rangle \alpha_2 \mid \dots \mid \langle A \rangle \alpha_m \mid \beta_1 \mid \beta_2 \mid \dots \mid \beta_n$$

they can be replaced by the following:

$$\begin{aligned}\langle A \rangle &::= \beta_1 \langle A' \rangle \mid \beta_2 \langle A' \rangle \mid \dots \mid \beta_n \langle A' \rangle \\ \langle A' \rangle &::= \alpha_1 \langle A' \rangle \mid \alpha_2 \langle A' \rangle \mid \dots \mid \alpha_m \langle A' \rangle \mid \varepsilon\end{aligned}$$

Example

Consider the simplified expression grammar:

$$\begin{aligned} E &::= E + T \mid T \\ T &::= \text{id} \mid \text{num} \end{aligned}$$

After eliminating left-recursion:

$$\begin{aligned} E &::= TE' \\ E' &::= +TE' \mid \epsilon \\ T &::= \text{id} \mid \text{num} \end{aligned}$$

Looking ahead to drive the choice of productions: $x + 5$.

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How much lookahead is needed?

We saw that top-down parsers need to select a production rule at every step, for which we may have to look ahead in the input string.

Do we need arbitrary lookahead to parse CFGs?

- in general, yes
- use the Earley or CYK algorithms

Fortunately

- large subclasses of CFGs can be parsed with limited lookahead
- most programming language constructs can be expressed in a grammar that falls in these subclasses

Among the interesting subclasses are:

- LL(1): left to right scan, left-most derivation, 1-token lookahead;
- and
- LR(1): left to right scan, reversed right-most derivation, 1-token lookahead

Recursive descent parsing and Predictive parsing

- If top-down parsing is performed recursively, it is also called *recursive descent parsing*.
 - To prevent infinite recursion, the grammar should not be left-recursive.
 - In general, may require backtracking if the wrong production rule is picked.
- Top-down parsing with lookahead which ensures that the correct production rule is always picked is called *predictive parsing*.

Recursive descent parsing

A set of procedures, one for each non-terminal.

```
1 int A()  
2 begin  
3   foreach production of the form  $A \rightarrow X_1X_2X_3 \cdots X_k$  do  
4     for  $i = 1$  to  $k$  do  
5       if  $X_i$  is a non-terminal then  
6         if  $(X_i() = 0)$  then  
7           backtrack; break; // Try the next production  
8       else if  $X_i$  matches the current input symbol  $a$  then  
9         advance the input to the next symbol;  
10      else  
11        backtrack; break; // Try the next production  
12      if  $i = k + 1$  then  
13        return 1; // Success  
14  return 0; // Failure
```

Recursive descent parsing

- Backtracks in general – in practise may not do much.
- How to backtrack?
- A left recursive grammar will lead to infinite loop.

Predictive parsing

Basic idea:

- For any two productions $A \rightarrow \alpha \mid \beta$, we would like a distinct way of choosing the correct production to expand.
- For some RHS $\alpha \in G$, define $\text{FIRST}(\alpha)$ as the set of tokens that appear first in some string derived from α .
 - That is, for some $a \in V_t$, $a \in \text{FIRST}(\alpha)$ iff. $\alpha \Rightarrow^* a\gamma$.

Key property:

- Whenever two productions $A \rightarrow \alpha$ and $A \rightarrow \beta$ both appear in the grammar, we would like:
 - $\text{FIRST}(\alpha) \cap \text{FIRST}(\beta) = \emptyset$
- This would allow the parser to make a correct choice with a lookahead of only one symbol!

Issue:

- If the grammar has two productions rules of the form $A \rightarrow \alpha\beta_1 \mid \alpha\beta_2$, we cannot directly use predictive parsing.

Left factoring

Some grammars can be transformed by left-factoring to enable predictive parsing.

For each non-terminal A find the longest prefix α common to two or more of its production rules.

if $\alpha \neq \varepsilon$ then replace all of the A productions

$$A \rightarrow \alpha\beta_1 \mid \alpha\beta_2 \mid \cdots \mid \alpha\beta_n$$

with

$$A \rightarrow \alpha A'$$

$$A' \rightarrow \beta_1 \mid \beta_2 \mid \cdots \mid \beta_n$$

where A' is a new non-terminal.

Repeat until no two alternatives for a single non-terminal have a common prefix.

Example

There are two non-terminals to left factor:

$$\begin{aligned}\langle \text{expr} \rangle &::= \langle \text{term} \rangle + \langle \text{expr} \rangle \\ &\quad | \langle \text{term} \rangle - \langle \text{expr} \rangle \\ &\quad | \langle \text{term} \rangle \\ \langle \text{term} \rangle &::= \langle \text{factor} \rangle * \langle \text{term} \rangle \\ &\quad | \langle \text{factor} \rangle / \langle \text{term} \rangle \\ &\quad | \langle \text{factor} \rangle\end{aligned}$$

Question: What's different here from the previous similar grammar that we've seen?

Applying the transformation:

$$\begin{aligned}\langle \text{expr} \rangle &::= \langle \text{term} \rangle \langle \text{expr}' \rangle \\ \langle \text{expr}' \rangle &::= + \langle \text{expr} \rangle \\ &\quad | - \langle \text{expr} \rangle \\ &\quad | \epsilon \\ \langle \text{term} \rangle &::= \langle \text{factor} \rangle \langle \text{term}' \rangle \\ \langle \text{term}' \rangle &::= * \langle \text{term} \rangle \\ &\quad | / \langle \text{term} \rangle \\ &\quad | \epsilon\end{aligned}$$

Left-recursion Elimination

- Predictive Parsing is a form of recursive-descent parsing, and hence cannot handle grammars with left recursion.
- We have seen how to eliminate immediate left-recursion, i.e. when there is a production rule of the form $A \rightarrow A\alpha$.
- However, left-recursion can also be indirect.
 - Example: $A \rightarrow B\alpha$ and $B \rightarrow A\beta$.
- In the general case, A grammar is left-recursive if $\exists A \in V_n$ such that $A \Rightarrow^+ A\alpha$ for some string α .

Indirect Left-recursion Elimination

Given a left-factored CFG, to eliminate left-recursion:

- 1 **Input:** Grammar G with no *cycles* (no $A \Rightarrow^* A$) and no ϵ productions.
- 2 **Output:** Equivalent grammar with no left-recursion.
- 3 **begin**
- 4 Arrange the non terminals in some order $A_1, A_2, \dots, A_n$;
- 5 **foreach** $i = 1 \dots n$ **do**
- 6 **foreach** $j = 1 \dots i - 1$ **do**
- 7 For production p of the form $A_i \rightarrow A_j \gamma$ and
 $A_j \rightarrow \delta_1 | \delta_2 | \dots | \delta_k$;
- 8 Replace the production p by:
- 9 $A_i \rightarrow \delta_1 \gamma | \delta_2 \gamma | \dots | \delta_k \gamma$;
- 10 Eliminate immediate left recursion in A_i ;

Example: eliminating indirect left recursion

Consider the grammar

$$\begin{aligned}\langle S \rangle &::= \langle A \rangle a \mid b \\ \langle A \rangle &::= \langle S \rangle d \mid c\end{aligned}$$

with indirect left recursion $\langle S \rangle \Rightarrow^* \langle S \rangle da$.

1. Order the nonterminals: $A_1 = \langle S \rangle$, $A_2 = \langle A \rangle$. For $i = 1$ there is no earlier nonterminal to substitute.

2. Process $i = 2, j = 1$: substitute $\langle S \rangle ::= \langle A \rangle a \mid b$ into $\langle A \rangle ::= \langle S \rangle d \mid c$:

$$\langle A \rangle ::= \langle A \rangle ad \mid bd \mid c$$

The indirect recursion is now immediate.

3. Eliminate the immediate left recursion in $\langle A \rangle$ ($\alpha = ad$, $\beta_1 = bd$, $\beta_2 = c$):

$$\begin{aligned}\langle S \rangle &::= \langle A \rangle a \mid b \\ \langle A \rangle &::= bd\langle A' \rangle \mid c\langle A' \rangle \\ \langle A' \rangle &::= ad\langle A' \rangle \mid \epsilon\end{aligned}$$

Indirect Left-recursion Elimination Algorithm Analysis

- At the end of i th iteration of the outer loop, the algorithm ensures that in all productions of the form $A_i \rightarrow A_j \gamma$, $i < j$.
- The algorithm assumes that the grammar has no cycles, i.e. $A \Rightarrow^* A$ is not possible for any non-terminal A .
- *Questions to ponder:*
 - What happens if there are cycles in the input grammar?
 - What happens if there are ε -productions in the input grammar?
- With a cycle, substitution can produce $A \rightarrow A$, and immediate left-recursion elimination does not work on it: the rewrite assumes $A \rightarrow A\alpha$ with $\alpha \neq \varepsilon$. Remove cycles first.

Why ε -productions break the algorithm

Order the nonterminals as $A_1 = A, A_2 = B, A_3 = C$:

$$A \rightarrow Ca \mid b$$

$$B \rightarrow \varepsilon$$

$$C \rightarrow BA$$

For $i = 3, j = 1$: $C \rightarrow BA$ does not begin with $A_1 = A$, so the algorithm makes no substitution.

Next, $j = 2$: substitute $B \rightarrow \varepsilon$:

$$C \rightarrow \varepsilon A = A.$$

But the loop does not revisit $j = 1$. Therefore

$$A \Rightarrow Ca \Rightarrow Aa,$$

and the indirect left recursion remains. Eliminate ε -productions before running the algorithm.

Which grammars does this leave out?

Both preconditions rule out grammars, so it is worth asking what they rule out in terms of *languages*.

- Does the algorithm work for all context-free languages?
 - Yes, it works for all CFL which do not contain ε . For any such CFL, we can always obtain a CFG which does not contain ε -productions and unit-productions.

Generality

Question:

By left factoring and eliminating left-recursion, can we transform an arbitrary context-free grammar to a form where it can be predictively parsed with a single token lookahead?

Answer: No. Example:

$$\{a^n 0 b^n \mid n \geq 1\} \cup \{a^n 1 b^{2n} \mid n \geq 1\}$$

A CFG is:

$$\begin{array}{ll} S \rightarrow A \mid B & A \rightarrow aAb \mid a0b \\ & B \rightarrow aBbb \mid a1bb \end{array}$$

- **Not LL(k), for any fixed k :** while choosing $S \rightarrow A$ or $S \rightarrow B$, lookahead may contain only a 's.

Where we are

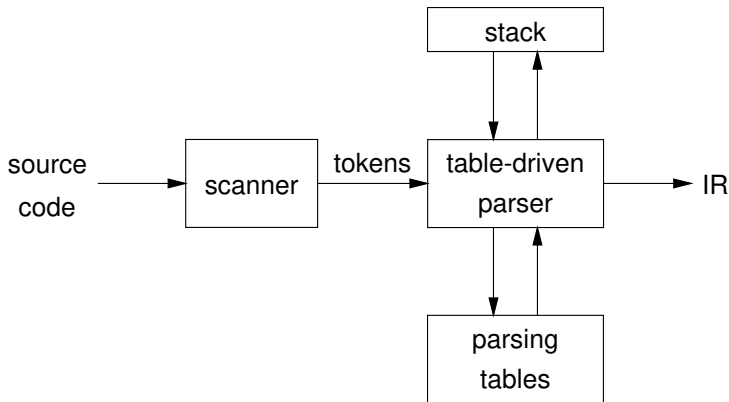
1 Context-free grammars

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Non-recursive predictive parsing

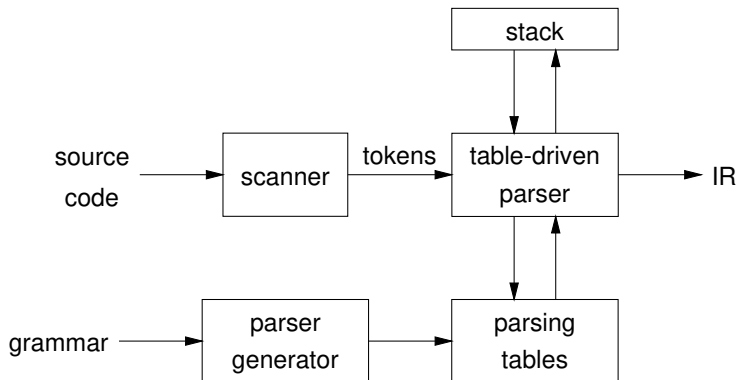
Now, a predictive parser looks like:



Rather than writing recursive code, we build tables.
Building tables can be automated easily.

Table-driven parsers

A parser generator system often looks like:



- We will first look at the information required for generating the parsing table.

FIRST

For a string of grammar symbols α , define $\text{FIRST}(\alpha)$ as:

- the set of terminals that begin strings derived from α :
 $\{a \in V_t \mid \alpha \Rightarrow^* a\beta\}$
- If $\alpha \Rightarrow^* \varepsilon$ then $\varepsilon \in \text{FIRST}(\alpha)$

To build $\text{FIRST}(X)$:

- ① If $X \in V_t$ then $\text{FIRST}(X)$ is $\{X\}$
- ② If $X \rightarrow \varepsilon$ then add ε to $\text{FIRST}(X)$
- ③ If $X \rightarrow Y_1 Y_2 \cdots Y_k$:
 - ① Put $\text{FIRST}(Y_1) - \{\varepsilon\}$ in $\text{FIRST}(X)$
 - ② $\forall i : 1 < i \leq k$, if $\varepsilon \in \text{FIRST}(Y_1) \cap \cdots \cap \text{FIRST}(Y_{i-1})$
(i.e., $Y_1 \cdots Y_{i-1} \Rightarrow^* \varepsilon$)
then put $\text{FIRST}(Y_i) - \{\varepsilon\}$ in $\text{FIRST}(X)$
 - ③ If $\varepsilon \in \text{FIRST}(Y_1) \cap \cdots \cap \text{FIRST}(Y_k)$ then put ε in $\text{FIRST}(X)$

Repeat until no more additions can be made.

FOLLOW

For a non-terminal A , define $\text{FOLLOW}(A)$ as
the set of terminals that can appear immediately to the right of A in some sentential form

We choose $A \rightarrow \alpha$ by what α starts with. A production that produces *nothing* has no such symbol:

$A \rightarrow \alpha$ look *inside*: $\text{FIRST}(\alpha)$
 $A \rightarrow \epsilon$ look *after*: $\text{FOLLOW}(A)$

Why FOLLOW: an example

$$\begin{array}{ll} E \rightarrow TE' & T \rightarrow FT' \\ E' \rightarrow +E \mid \varepsilon & T' \rightarrow *T \mid /T \mid \varepsilon \end{array}$$

We are at T' and the lookahead is $+$:

$$\begin{array}{ll} T' \rightarrow *T & \text{needs } * \\ T' \rightarrow /T & \text{needs } / \\ T' \rightarrow \varepsilon & \text{needs } \textit{nothing} \end{array}$$

T' ends a T , a T is followed by E' , and E' can start with $+$. So $+\in\text{FOLLOW}(T')$: T' vanishes here, and the $+$ is matched by E' further out.

Building FOLLOW

A terminal symbol has no FOLLOW set.

To build $\text{FOLLOW}(A)$:

- ① Put $\$$ in $\text{FOLLOW}(\langle \text{goal} \rangle)$
- ② If $A \rightarrow \alpha B \beta$:
 - ① Put $\text{FIRST}(\beta) - \{\epsilon\}$ in $\text{FOLLOW}(B)$
 - ② If $\beta = \epsilon$ (i.e., $A \rightarrow \alpha B$) or $\epsilon \in \text{FIRST}(\beta)$ (i.e., $\beta \Rightarrow^* \epsilon$) then put $\text{FOLLOW}(A)$ in $\text{FOLLOW}(B)$

Repeat until no more additions can be made

LL(1) grammars

Previous definition (slide 44)

A grammar G is LL(1) iff. for all non-terminals A , each distinct pair of productions $A \rightarrow \beta$ and $A \rightarrow \gamma$ satisfy the condition $\text{FIRST}(\beta) \cap \text{FIRST}(\gamma) = \phi$.

What if $\epsilon \in \text{FIRST}(\beta)$?

Consider that the current input symbol is a . Choosing $A \rightarrow \beta$ consumes nothing, so it is the right move exactly when a can legally come after A , i.e. when $a \in \text{FOLLOW}(A)$. That introduces ambiguity between choosing:

- $A \rightarrow \beta$ when $a \in \text{FOLLOW}(A)$
- $A \rightarrow \gamma$ when $a \in \text{FIRST}(\gamma)$

Ambiguity is bad because we may need to backtrack: not predictive parsing anymore!

LL(1) grammars

Revised definition

A grammar G is LL(1) iff. for each set of productions $A \rightarrow \alpha_1 \mid \alpha_2 \mid \dots \mid \alpha_n$:

- ① *FIRST(α_1), FIRST(α_2), ..., FIRST(α_n) are all pairwise disjoint*
- ② *If $\alpha_i \Rightarrow^* \varepsilon$ then
FIRST(α_j) $\cap$ FOLLOW(A) = ϕ , $\forall 1 \leq j \leq n, i \neq j$.*

If G is ε -free, condition 1 is sufficient.

LL(1) grammars

Provable facts about LL(1) grammars:

① No left-recursive grammar is LL(1)

- Direct case: $A \rightarrow A\alpha \mid \beta$ (useful productions).
- Case 1: β is non-nullable. For any $a \in \text{FIRST}(\beta)$,

$$A\alpha \Rightarrow \beta\alpha \Rightarrow^* a\gamma\alpha.$$

Thus $a \in \text{FIRST}(A\alpha)$ too: condition 1 fails.

- Case 2: β is nullable, so A is nullable.
 - If α is nullable, then $\varepsilon \in \text{FIRST}(\beta) \cap \text{FIRST}(A\alpha)$: condition 1 fails.
 - Otherwise, for any $a \in \text{FIRST}(\alpha)$,

$$\begin{array}{llll} A\alpha & \Rightarrow^* & \alpha \Rightarrow^* a\gamma & \Rightarrow a \in \text{FIRST}(A\alpha), \\ A & \Rightarrow & A\alpha \Rightarrow^* Aa\gamma & \Rightarrow a \in \text{FOLLOW}(A). \end{array}$$

The same a triggers both alternatives: condition 2 fails.

- Indirect case: expand the cycle to expose $A \Rightarrow^+ A\alpha$; the same case analysis applies.

LL(1) grammars

Provable facts about LL(1) grammars:

- ② No ambiguous grammar is LL(1)
 - Some string has two leftmost derivations. Where they first differ, one A is expanded two ways and both still reach that string, so their FIRST (or FIRST/FOLLOW) sets overlap. Hence LL(1) $\Rightarrow$ unambiguous, though not conversely.

LL(1) grammars

Provable facts about LL(1) grammars:

- ③ Some languages have no LL(1) grammar
 - Some CFLs are inherently ambiguous i.e., no unambiguous CFGs exist for that CFL.

- ④ A grammar which is not LL(1) may be converted into a LL(1) grammar.
 - Consider $S \rightarrow aS \mid a$. Not LL(1) since $\text{FIRST}(aS) = \text{FIRST}(a)$. Use left-factoring to get:
$$S \rightarrow aS'$$
$$S' \rightarrow aS' \mid \epsilon$$
accepts the same language and is LL(1)

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LL(1) parse table construction

Input: Grammar G

Output: Parsing table M

Method:

① $\forall$ productions $A \rightarrow \alpha$:

① $\forall a \in \text{FIRST}(\alpha)$, add $A \rightarrow \alpha$ to $M[A, a]$

② If $\epsilon \in \text{FIRST}(\alpha)$:

① $\forall b \in \text{FOLLOW}(A)$, add $A \rightarrow \alpha$ to $M[A, b]$

② If $\$ \in \text{FOLLOW}(A)$ then add $A \rightarrow \alpha$ to $M[A, \$]$

② Set each undefined entry of M to `error`

If $\exists M[A, a]$ with multiple entries then grammar is not LL(1).

Example

Our expression grammar:

1. $S \rightarrow E$
2. $E \rightarrow TE'$
3. $E' \rightarrow +E$
4. $\quad \quad \quad | -E$
5. $\quad \quad \quad | \epsilon$
6. $T \rightarrow FT'$
7. $T' \rightarrow *T$
8. $\quad \quad \quad | /T$
9. $\quad \quad \quad | \epsilon$
10. $F \rightarrow \text{num}$
11. $\quad \quad \quad | \text{id}$

	FIRST	FOLLOW	id	num	+	-	*	/	\$
S									
E									
E'									
T									
T'									
F									
id									
num									
*									
/									
+									
-									

Example: Calculating FIRST

1.	S	$\rightarrow E$	6.	T	$\rightarrow FT'$
2.	E	$\rightarrow TE'$	7.	T'	$\rightarrow *T$
3.	E'	$\rightarrow +E$	8.		$\mid /T$
4.		$\mid -E$	9.		$\mid \epsilon$
5.		$\mid \epsilon$	10.	F	$\rightarrow \text{num}$
			11.		$\mid \text{id}$

$\text{FIRST}(E)$	$\subseteq$	$\text{FIRST}(S)$	From 1
$\text{FIRST}(T)$	$\subseteq$	$\text{FIRST}(E)$	From 2
$\{+, -, \epsilon\}$	$\subseteq$	$\text{FIRST}(E')$	From 3–5
$\text{FIRST}(F)$	$\subseteq$	$\text{FIRST}(T)$	From 6
$\{*, /, \epsilon\}$	$\subseteq$	$\text{FIRST}(T')$	From 7–9
$\{\text{num}, \text{id}\}$	$\subseteq$	$\text{FIRST}(F)$	From 10, 11

Example: Calculating FIRST

- | | | | | | |
|----|------|-------------------|-----|------|--------------------------|
| 1. | S | $\rightarrow E$ | 6. | T | $\rightarrow FT'$ |
| 2. | E | $\rightarrow TE'$ | 7. | T' | $\rightarrow *T$ |
| 3. | E' | $\rightarrow +E$ | 8. | | $\mid /T$ |
| 4. | | $\mid -E$ | 9. | | $\mid \epsilon$ |
| 5. | | $\mid \epsilon$ | 10. | F | $\rightarrow \text{num}$ |
| | | | 11. | | $\mid \text{id}$ |

	FIRST	FOLLOW	id	num	+	-	*	/	\$
S	num,id								
E	num,id								
E'	$\epsilon, +, -$								
T	num,id								
T'	$\epsilon, *, /$								
F	num,id								
id	id	-							
num	num	-							
*	*	-							
/	/	-							
+	+	-							
-	-	-							

Example: Calculating FOLLOW

1.	S	$\rightarrow E$	6.	T	$\rightarrow FT'$
2.	E	$\rightarrow TE'$	7.	T'	$\rightarrow *T$
3.	E'	$\rightarrow +E$	8.		$ /T$
4.		$ -E$	9.		$ \epsilon$
5.		$ \epsilon$	10.	F	$\rightarrow \text{num}$
			11.		$ \text{id}$

$\{\$ \}$	$\subseteq$	$\text{FOLLOW}(S)$	Start-symbol rule
$\text{FOLLOW}(S)$	$\subseteq$	$\text{FOLLOW}(E)$	From 1
$\text{FIRST}(E') - \{\epsilon\}$	$\subseteq$	$\text{FOLLOW}(T)$	From 2
$\text{FOLLOW}(E)$	$\subseteq$	$\text{FOLLOW}(E')$	From 2
$\text{FOLLOW}(E)$	$\subseteq$	$\text{FOLLOW}(T)$	From 2
$\text{FOLLOW}(E')$	$\subseteq$	$\text{FOLLOW}(E)$	From 3, 4
$\text{FIRST}(T') - \{\epsilon\}$	$\subseteq$	$\text{FOLLOW}(F)$	From 6
$\text{FOLLOW}(T)$	$\subseteq$	$\text{FOLLOW}(T')$	From 6
$\text{FOLLOW}(T)$	$\subseteq$	$\text{FOLLOW}(F)$	From 6
$\text{FOLLOW}(T')$	$\subseteq$	$\text{FOLLOW}(T)$	From 7, 8

Example: Calculating FOLLOW

1. $S \rightarrow E$
2. $E \rightarrow TE'$
3. $E' \rightarrow +E$
4. $\quad \quad \quad | -E$
5. $\quad \quad \quad | \epsilon$
6. $T \rightarrow FT'$
7. $T' \rightarrow *T$
8. $\quad \quad \quad | /T$
9. $\quad \quad \quad | \epsilon$
10. $F \rightarrow \text{num}$
11. $\quad \quad \quad | \text{id}$

	FIRST	FOLLOW	id	num	+	-	*	/	\$
S	num,id	\$							
E	num,id	\$							
E'	$\epsilon, +, -$	\$							
T	num,id	$+, -, \$$							
T'	$\epsilon, *, /$	$+, -, \$$							
F	num,id	$+, -, *, /, \$$							
id	id	-							
num	num	-							
*	*	-							
/	/	-							
+	+	-							
-	-	-							

Example: Calculating the Parsing Table

- | | | | | | |
|----|------|-------------------|-----|------|--------------------------|
| 1. | S | $\rightarrow E$ | 6. | T | $\rightarrow FT'$ |
| 2. | E | $\rightarrow TE'$ | 7. | T' | $\rightarrow *T$ |
| 3. | E' | $\rightarrow +E$ | 8. | | $\mid /T$ |
| 4. | | $\mid -E$ | 9. | | $\mid \epsilon$ |
| 5. | | $\mid \epsilon$ | 10. | F | $\rightarrow \text{num}$ |
| | | | 11. | | $\mid \text{id}$ |

	FIRST	FOLLOW	id	num	+	-	*	/	\$
S	num,id	\$	1	1	-	-	-	-	-
E	num,id	\$	2	2	-	-	-	-	-
E'	$\epsilon, +, -$	\$	-	-	3	4	-	-	5
T	num,id	$+, -, \$$	6	6	-	-	-	-	-
T'	$\epsilon, *, /$	$+, -, \$$	-	-	9	9	7	8	9
F	num,id	$+, -, *, /, \$$	11	10	-	-	-	-	-
id	id	-							
num	num	-							
*	*	-							
/	/	-							
+	+	-							
-	-	-							

Table driven Predictive parsing

Input: A string w and a parsing table M for a grammar G

Output: If w is in $L(G)$, a leftmost derivation of w ; otherwise, indicate an error

```
1 push $ onto the stack; push  $S$  onto the stack;
2 let  $a = \text{first\_symbol}(w)$ ;
3  $X = \text{stack.top}()$ ;
4 while  $X \neq \$$  do
5     if  $X == a$  then
6          $\text{stack.pop}()$ ; let  $a = \text{next\_symbol}(w)$ ;
7     else if  $X$  is a terminal then
8          $\text{error}()$ ;
9     else if  $M[X, a]$  is an error entry then
10         $\text{error}()$ ;
11    else if  $M[X, a] = X \rightarrow Y_1 Y_2 \cdots Y_k$  then
12        output the production  $X \rightarrow Y_1 Y_2 \cdots Y_k$ ;
13         $\text{stack.pop}()$ ;
14        push  $Y_k, Y_{k-1}, \cdots Y_1$  in that order;
15     $X = \text{stack.top}()$ ;
```

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A grammar that is not LL(1): the dangling else

$$\begin{aligned}\langle \text{stmt} \rangle &::= \text{if } \langle \text{expr} \rangle \text{ then } \langle \text{stmt} \rangle \\ &\quad | \text{if } \langle \text{expr} \rangle \text{ then } \langle \text{stmt} \rangle \text{ else } \langle \text{stmt} \rangle \\ &\quad | \text{other}\end{aligned}$$

Left-factored:

1. $\langle \text{stmt} \rangle ::= \text{if } \langle \text{expr} \rangle \text{ then } \langle \text{stmt} \rangle \langle \text{stmt}' \rangle$
2. $\quad \quad \quad | \text{other}$
3. $\langle \text{stmt}' \rangle ::= \text{else } \langle \text{stmt} \rangle$
4. $\quad \quad \quad | \epsilon$

Constraints relevant to the conflict (others omitted):

$\text{FIRST}(\langle \text{stmt}' \rangle)$	$= \{\text{else}, \epsilon\}$	From 3, 4
$\$$	$\in \text{FOLLOW}(\langle \text{stmt} \rangle)$	Start-symbol rule
$\text{FOLLOW}(\langle \text{stmt} \rangle)$	$\subseteq \text{FOLLOW}(\langle \text{stmt}' \rangle)$	From 1
$\text{FIRST}(\langle \text{stmt}' \rangle) - \{\epsilon\}$	$\subseteq \text{FOLLOW}(\langle \text{stmt} \rangle)$	From 1
$\text{FOLLOW}(\langle \text{stmt}' \rangle)$	$\subseteq \text{FOLLOW}(\langle \text{stmt} \rangle)$	From 3

A grammar that is not LL(1): the dangling else

Left-factored: $\langle \text{stmt} \rangle ::= \text{if } \langle \text{expr} \rangle \text{ then } \langle \text{stmt} \rangle \langle \text{stmt}' \rangle \mid \text{other}$
 $\langle \text{stmt}' \rangle ::= \text{else } \langle \text{stmt} \rangle \mid \epsilon$

Picking the smallest set that can satisfy the constraints gives us:

$$\text{FOLLOW}(\langle \text{stmt}' \rangle) = \{\text{else}, \$\}$$

Given $\langle \text{stmt}' \rangle \Rightarrow^* \epsilon$, LL(1) grammar requires

$$\text{FIRST}(\text{else} \langle \text{stmt} \rangle) \cap \text{FOLLOW}(\langle \text{stmt}' \rangle) = \emptyset.$$

$$\text{But } \text{FIRST}(\text{else} \langle \text{stmt} \rangle) \cap \text{FOLLOW}(\langle \text{stmt}' \rangle) = \{\text{else}\}$$

The parsing table entry for $M[\langle \text{stmt}' \rangle, \text{else}]$ will contain both:

- $\langle \text{stmt}' \rangle ::= \text{else} \langle \text{stmt} \rangle$
- $\langle \text{stmt}' \rangle ::= \epsilon$

Intuitively, prioritise $\langle \text{stmt}' \rangle ::= \text{else} \langle \text{stmt} \rangle$ to associate `else` with closest `then`.

Another common example

- Here is a typical example where a programming language fails to be LL(1):

$$\begin{aligned}\langle \text{stmt} \rangle &\rightarrow \langle \text{assignment} \rangle \mid \langle \text{call} \rangle \mid \langle \text{other} \rangle \\ \langle \text{assignment} \rangle &\rightarrow \langle \text{id} \rangle = \langle \text{expr} \rangle \\ \langle \text{call} \rangle &\rightarrow \langle \text{id} \rangle (\langle \text{expr-list} \rangle)\end{aligned}$$

- This grammar is not in a form that can be left factored. We must first replace assignment and call by the right-hand sides of their defining productions:

$$\langle \text{stmt} \rangle \rightarrow \langle \text{id} \rangle = \langle \text{expr} \rangle \mid \langle \text{id} \rangle (\langle \text{expr-list} \rangle) \mid \langle \text{other} \rangle$$

- We left factor:

$$\begin{aligned}\langle \text{stmt} \rangle &\rightarrow \langle \text{id} \rangle \langle \text{stmt}' \rangle \mid \langle \text{other} \rangle \\ \langle \text{stmt}' \rangle &\rightarrow = \langle \text{expr} \rangle \mid (\langle \text{expr-list} \rangle)\end{aligned}$$

- See how the grammar obscures the language semantics.
 - Most of PL syntax cannot be expressed naturally as LL(1) grammar.

Error recovery in predictive parsing

- An error is detected on a terminal mismatch or when $M[A, a]$ is empty.
- **Panic mode:** discard input until a synchronizing token is reached.
 - If $a \in \text{FOLLOW}(A)$, then a can legally come after A . Report a missing or malformed A , pop A , and **do not consume** a .

Recovery example: $\text{id} * + \text{id}$

After matching $*$, the stack expects T , but the lookahead is $+$:

Stack	:	$\$$	E'	T
Input	:	$+$	id	$\$$

Therefore $M[T, +]$ is empty: the error is detected at $+$.

Since $+\in\text{FOLLOW}(T)$, report “expected an operand after $*$ ”, pop T , and continue at $+$.

The input remains invalid; recovery only lets parsing continue.